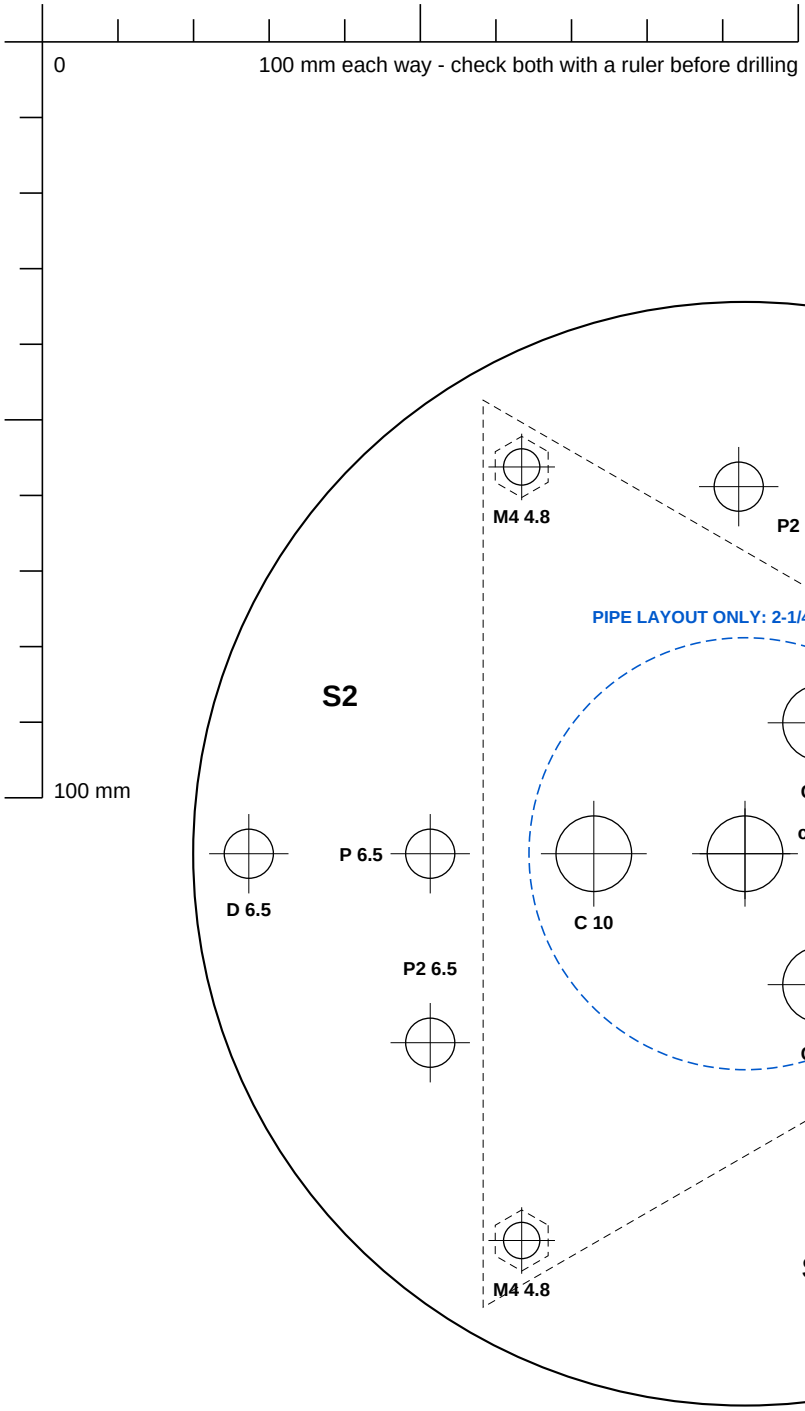
Shield deck drill template - 146 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (electronics side). Tape to the board, centre-punch each crosshair, then drill.



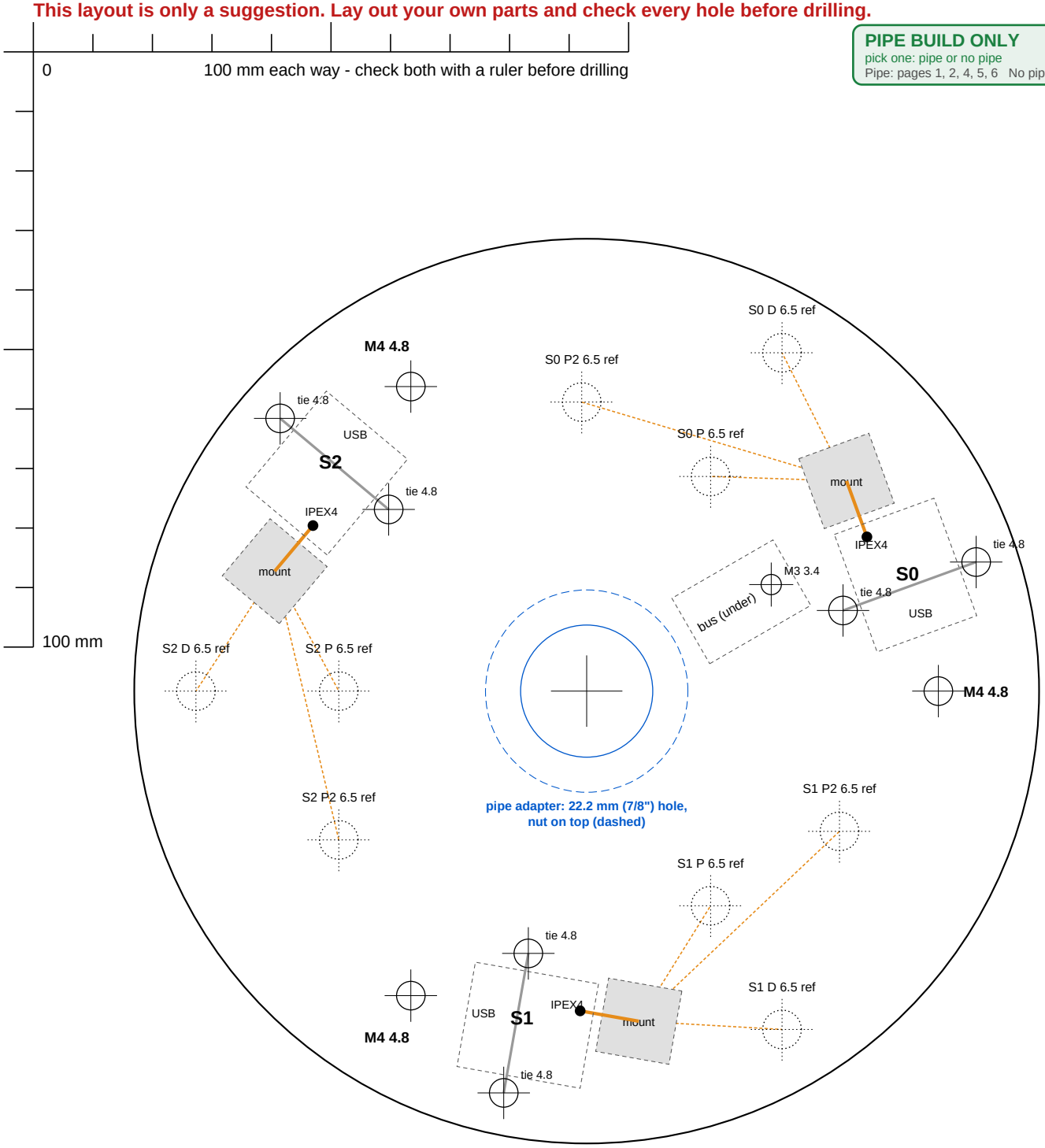
Solid circle = cut line (146 mm). Dashed triangle = where the 120 mm back panels stand, copper side out.
D 6.5 mm = rubber duck / monopole / dipole whip (31 mm in front of face) P / P2 6.5 mm = patch (7 mm in front of face; P2 25 mm off centre)
C 10 mm = dipole coax pass-through (inside core) M4 4.8 mm = bolts to the plate, nut flush in each core corner
Core 10 mm at the centre = antenna coax into the core; open it up with a hole saw (centre mark) if needed.
Blue dashed 57.1 mm (2-1/4" hole saw) = pipe layout only (page 2): the pipe passes into the core;
it removes the C holes. Other layouts keep the 10 mm core hole. EU: a 57 mm hole saw is the same size.
Arrow points to Sector 0. Holes are symmetric, so the template also works flipped (labels read mirrored).
Drawn the same way round as the plates: plate 12 o'clock up, S0 arrow 30 deg clockwise of it.

PIPE electronics plate - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. XIAOs, mounts + bus hang UNDER the plate (dashed); PoE splitter + ESP32 inside the core.

This layout is only a suggestion. Lay out your own parts and check every hole before drilling.

PIPE BUILD ONLY
pick one: pipe or no pipe
Pipe: pages 1, 2, 4, 5, 6 No pipe: pages 1, 3, 4, 6



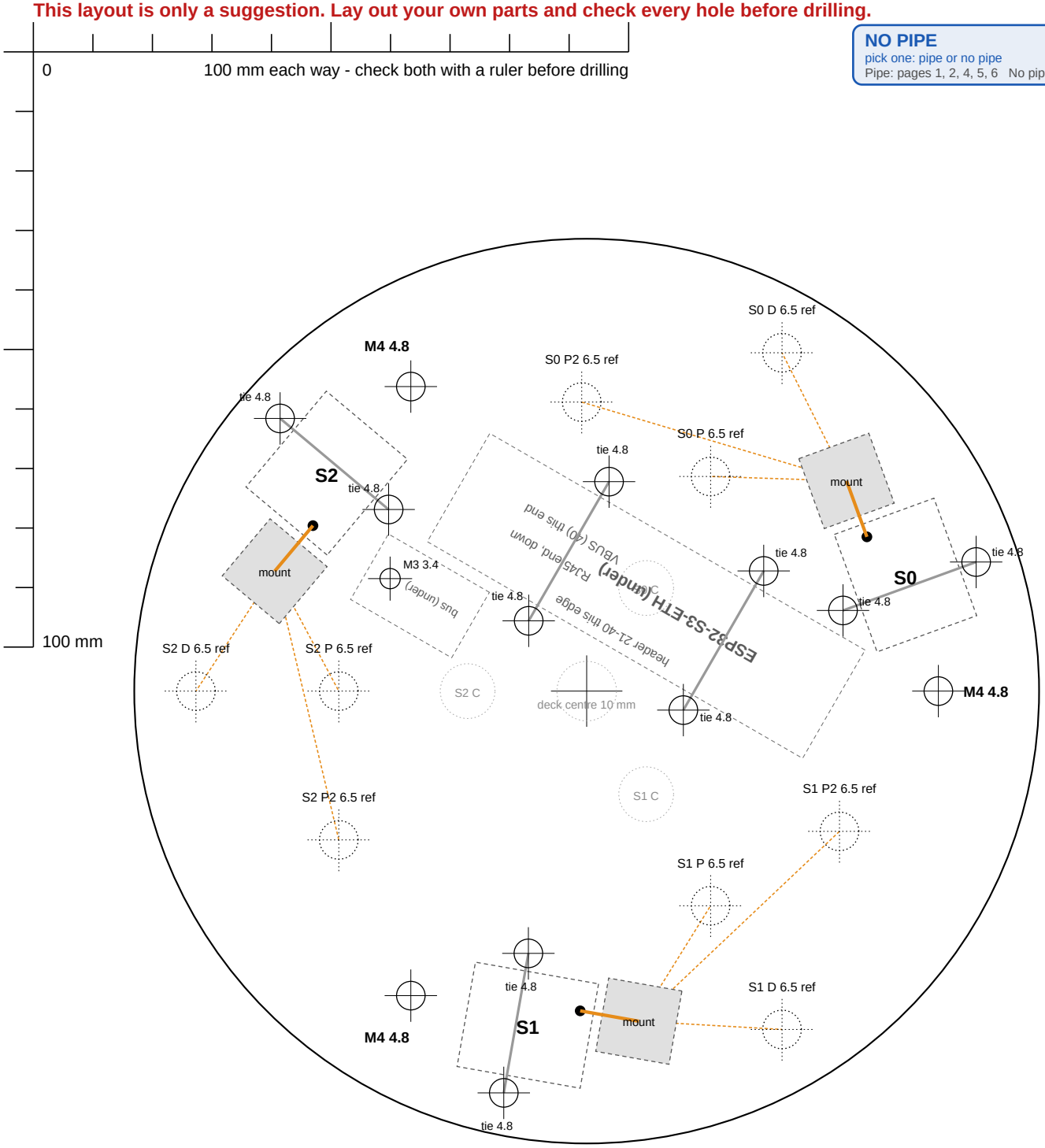
Hanging from the ceiling, top down: nut, enclosure top, this module plate, modules + wiring under it, deck, core.
Drill: 3 x 4.8 mm (M4) bolts to the deck, on the deck's M4 holes; 3.4 mm (M3) bus board;
tie hole pairs 4.8 mm (3.0-4.8 fits a 2.5 mm tie), one per XIAO. No pigtail or feed holes: the parts hang under the plate.
Antenna holes (dotted, 9 x 6.5 mm): the deck's D / P / P2 jacks, shown for reference only. Not by design; drill if you like.
Centre (blue): 22.2 mm (7/8") hole for the 1/2" male adapter's thread, which also goes through the
enclosure top; the nut on top (dashed, size est.) clamps both. Ethernet comes down the pipe (page 5).
EU: a 20 mm x 1/2" BSP male adaptor (R 1/2, 21 mm thread) fits the same 22 mm hole, held by a 1/2" BSP backnut.
Each XIAO (under the plate, component side down, beside its own sector): IPEX4 dot -> orange flat coax run ->
1/2" HS-100 tie mount (grey), which clamps it, then (dashed orange) down to whichever of its sector's D / P / P2
jacks is in use, straight below its hole. 5V, GND, D7 to the bus beside them; D6 through the deck's centre hole to
its ESP32 RX pin, beside TX, GND, VBUS coming up. Seen from the top the IPEX4 is on the left (the board is flipped).
OPTIONS (builder's choice): this pipe plate or the no-pipe split plate (page 3); bus board under the plate by the adapter,
or on the pipe inside the core (page 5: then skip the bus M3).

NO-PIPE electronics plate (split) - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. Radios, tie mounts, ESP32 + bus all hang UNDER the plate (dashed, seen through it).

This layout is only a suggestion. Lay out your own parts and check every hole before drilling.

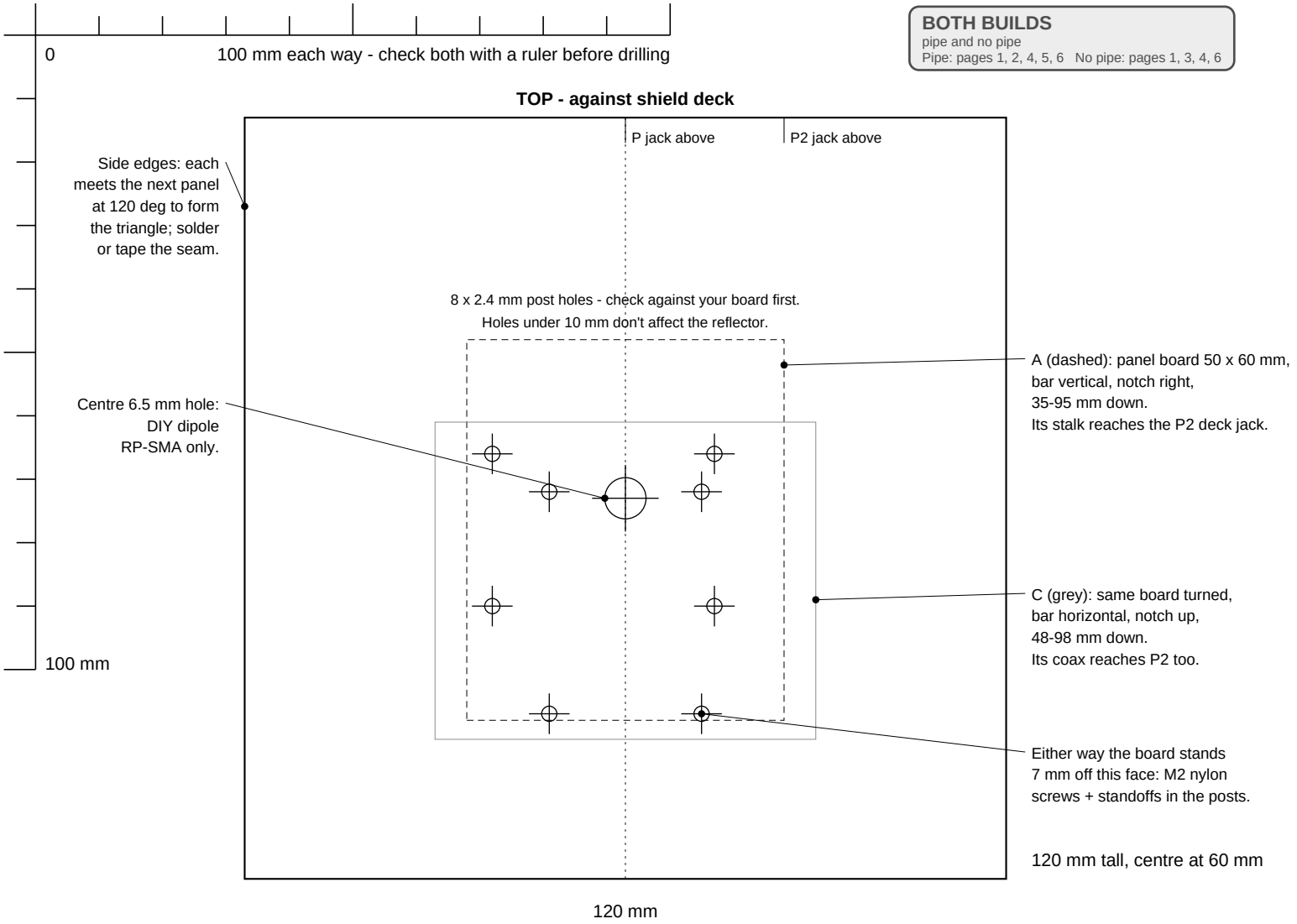
NO PIPE
pick one: pipe or no pipe
Pipe: pages 1, 2, 4, 5, 6 No pipe: pages 1, 3, 4, 6



Hanging from the ceiling, top down: enclosure top, this module plate, everything else under it, deck, core.
Drill: 3 x 4.8 mm (M4) bolts to the deck, on the deck's M4 holes; 3.4 mm (M3) bus board. No harness holes.
Tie hole pairs 4.8 mm (any 3.0-4.8 mm fits a 2.5 mm tie): one across each XIAO, two across the ESP32.
Antenna holes (dotted, 9 x 6.5 mm): the deck's D / P / P2 jacks, shown for reference only. Not by design; drill if you like.
XIAOs (in the gaps between the sectors' holes) on VHB + a zip tie (grey bar), IPEX4 dot -> orange coax -> 1/2" HS-100 tie mount, which clamps it, then on (dashed orange) to its own sector's D / P / P2 jack in use: one smooth sweep, bending down through the spacer gap. Nothing sits within 8 mm of an antenna hole (the coax rises straight from its jack).
ESP32: across the middle, RJ45 / USB-C end toward 10 o'clock, hung RJ45 side down; USB-C then faces the plate: standoffs or a slim right-angle plug. A 90 deg adapter turns the RJ45 sideways; patch + USB-C lead to the PoE splitter's 4" box.
Bus by the ESP32's RJ45 end (VBUS 40). Spacers 25-30 mm. Clearances are tight (~2-3 mm): check with the real parts.
Positions: parameters.csv (lay out in plate_layout.svg). The ESP32 only clears the deck's 10 mm centre hole: no hole saw.
OPTIONS (builder's choice): this no-pipe split plate or the pipe plate (page 2). Wiring: page 6.

Back panel (reflector face) template - 120 x 120 mm - PRINT AT 100%

Make 3. Viewed from the FRONT (copper side, facing the antenna). Top edge goes against the shield deck.



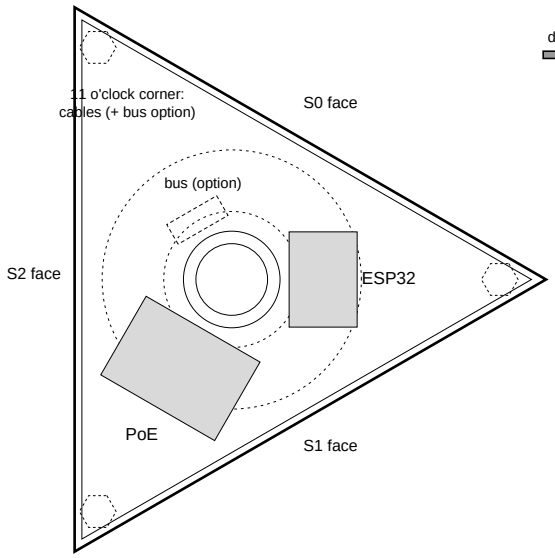
Whips (rubber duck / monopole / dipole) don't mount on this face: they hang on the deck's D holes, 31 mm in front of it (page 1).

Pipe stack - top and side views - scale 0.6 (NOT a drilling template)

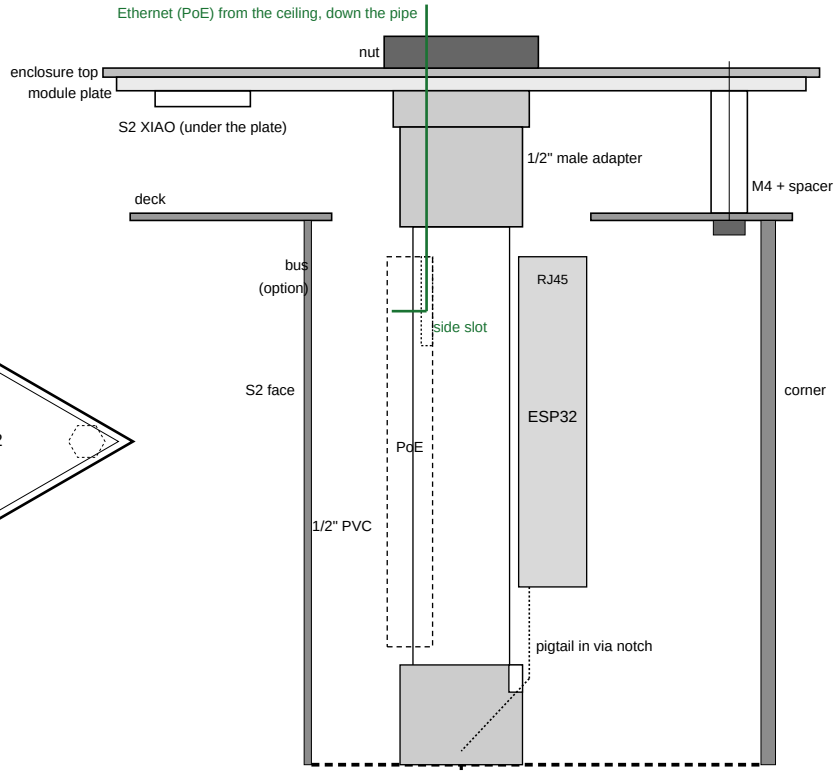
Hanging from the ceiling, top down: nut, enclosure top, module plate, modules, deck, core, ESP antenna. Frame: page 2.

PIPE BUILD ONLY
pick one: pipe or no pipe
Pipe: pages 1, 2, 4, 5, 6 No pipe: pages 1, 3, 4, 6

TOP: inside the core, looking down



SIDE: cut through 3 - 9 o'clock



1/2" cap, drilled and tapped for the RP-SMA

copper end plate (optional): bond all 3 edges or leave it off

dotted: adapter hex (above the deck)
and the 2-1/4" deck hole;
dotted hexes: M4 nuts under the deck

END PLATE: leave it off, or bond it all round - never
held by the antenna screw alone.
Worst: plate on the antenna screw only. Its loose edges make slots (a face is 120 mm; ~62 mm resonates at 2.4 GHz) that re-radiate and can detune the antennas.
Fine: no plate. The faces still shield the sides; the open end leaks ESP32 / splitter noise downward.
Best: plate bonded along all 3 edges, gaps < ~25 mm (copper tape over the seams, or spring tabs / screws every 20-25 mm), star washer under the antenna nut.
No plate: use a self-contained antenna on the cap (a sleeve-dipole rubber duck), not a bare 1/4-wave whip. Build without it first; add a bonded plate only if the XIAOs see noise with the ESP32 and splitter running.

OPTION: 1/2" EMT (steel conduit) instead of the PVC
EMT: 17.9 mm OD, 15.8 mm bore (PVC: 21.3 / 15.8 mm). Set-screw or compression connector + locknut in the same 7/8" (22.2 mm) hole; the deck hole stays.
RJ45: a crimped head only just fits the 15.8 mm bore (check the connector's throat too). Make sure there is NO boot or crimp sleeve on the plug going through.
Bond the tube to the deck all round where it passes: copper collar, ground clamp + braid, or conductive copper tape (solder won't take on galvanized steel).
ESP antenna: plastic cap or bushing, not bare steel. Bushing on every cut or drilled hole a cable passes near or through (side slot, tube ends), even deburred.
Outdoors: seal the zinc-copper bond (corrosion).
The drawing shows the PVC pipe.

ESP antenna (RP-SMA in the cap)

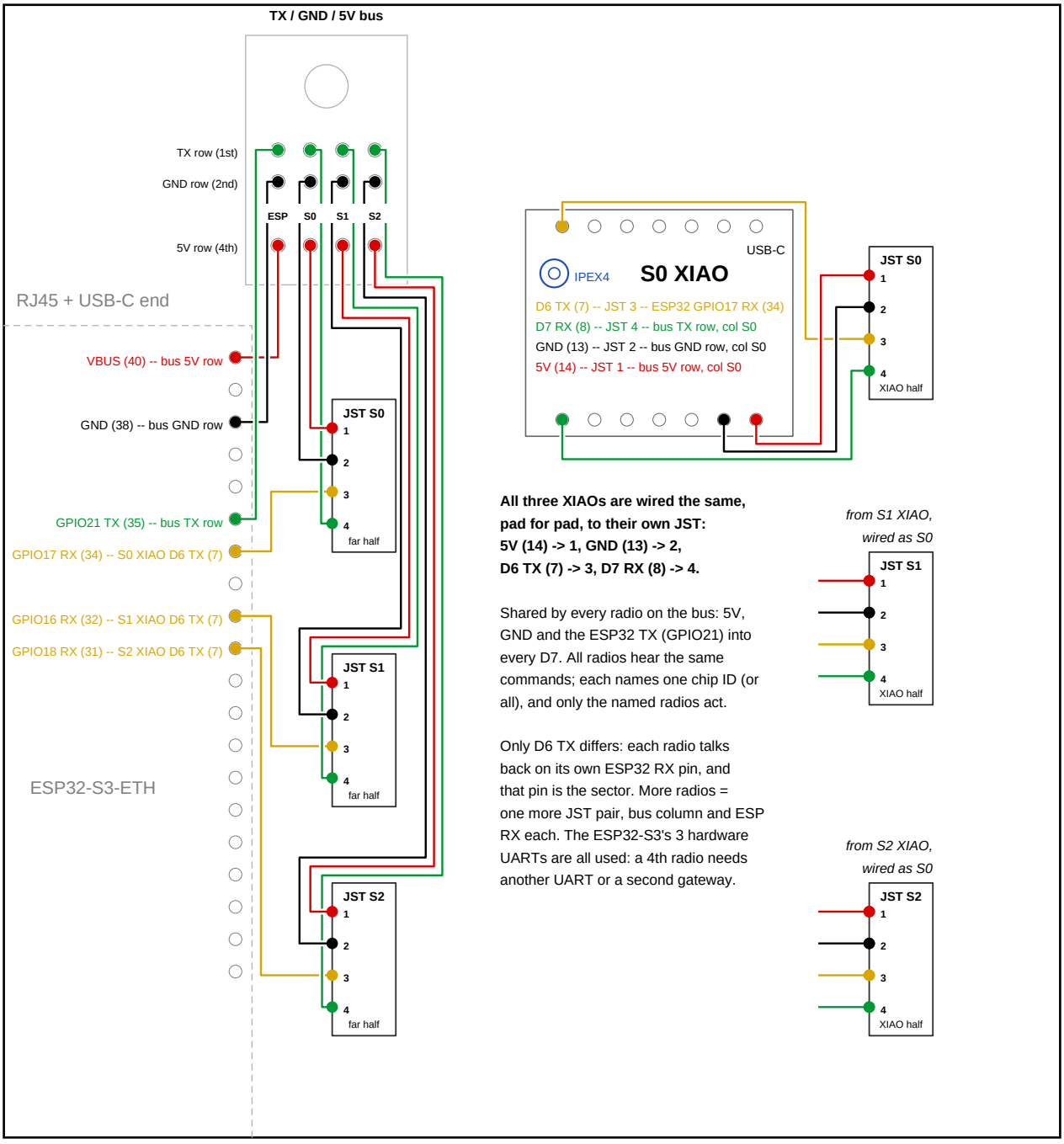
Pipe: 1/2" PVC (21.3 mm OD) from the male adapter's socket through the 2-1/4" deck hole, ~137 mm long (check the socket depth), capped flush with the core's far end. The cap is drilled and tapped: the ESP antenna's RP-SMA bulkhead threads into it, and the U.FL pigtail from the ESP32 enters through a notch in the cap rim. The antenna, screwed onto the bulkhead, clamps the copper end plate that closes the core (optional; see END PLATE, left).
Splitter and ESP32 zip-tied to the pipe facing the 3 and 7 o'clock corners (widest room), RJ45 / USB-C end up; the 11 o'clock corner is left for cables (and the bus option). Ethernet (PoE, green): from the ceiling down the adapter's bore and the pipe, out a side slot (about 12 x 16 mm, cut before gluing) beside the splitter. An RJ45 plug (11.7 mm wide) fits the ~16 mm bore: pull it through before fitting any boot, or crimp after. The adapter's thread goes up through the module plate and the enclosure top; the nut on top clamps both. Modules and the bus hang under the plate (page 2).
OPTION: the bus board can go on the pipe (dashed, 11 o'clock, opposite the ESP32 and splitter) instead of on the plate.
Side: dashed box = PoE splitter, behind the cut. XIAOs, tie mounts: as page 2. Estimates: adapter, nut, enclosure top, splitter size, board depths, antenna length.
EU parts: 20 mm PVC-U pressure pipe (20 mm OD, 1.3 mm under 1/2" Sch 40; same holes), a 20 mm solvent-weld end cap, a 20 mm x 1/2" BSP male adaptor and a 1/2" BSP backnut, and a 57 mm hole saw for the deck. The RP-SMA bulkhead's thread is 1/4"-36 UNS everywhere: if that tap is hard to find, drill the cap 6.5 mm and use the bulkhead's own nut.

50 mm at drawing scale

BOTH BUILDS
pipe and no pipe
Pipe: pages 1, 2, 4, 5, 6 No pipe: pages 1, 3, 4, 6

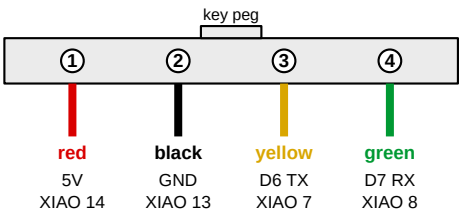
Wiring diagram - XIAO radios, JST pairs, bus board, ESP32

Not to scale for mounting: shows connections only. Pin numbers in brackets. Pad positions approximate.
JST halves drawn apart: XIAO half beside the radio, far half by the bus; same name = they plug together.



D6 -> its own ESP RX (the pin is the sector): S0 GPIO17 (34), S1 GPIO16 (32), S2 GPIO18 (31).
ESP TX GPIO21 (35) -> bus TX row -> every D7. Feed: TX (35) -> TX row, VBUS (40) -> 5V row, GND (38) -> GND row.
Bus: hole end up, rows bridged, 3rd row spare. ESP32 as drawn (RJ45 side up): VBUS (40) is the top right header pin.

JST head (either half), crimp side, key peg up:



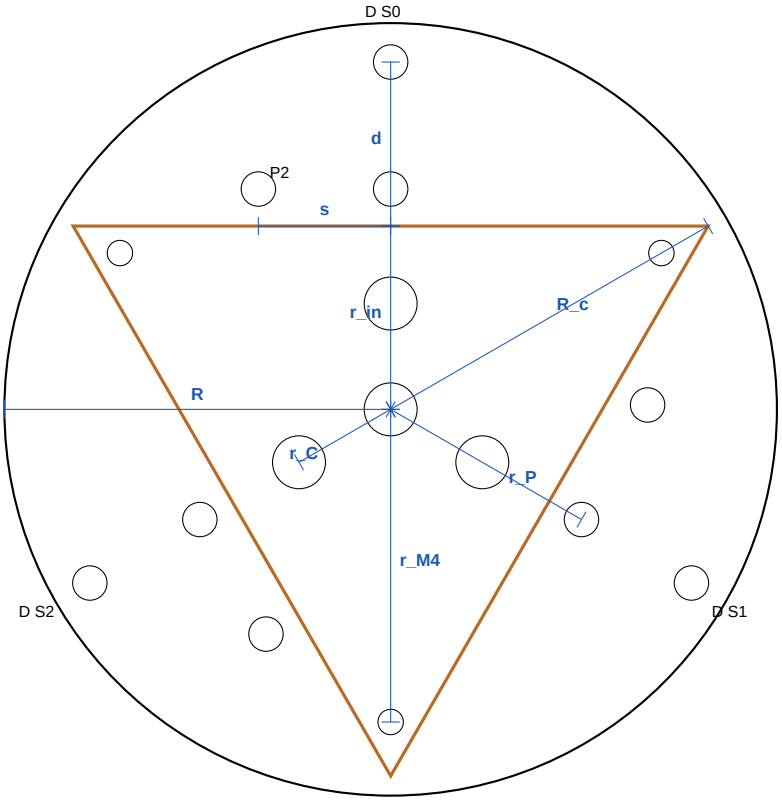
Pins left to right: 1 red 5V, 2 black GND, 3 yellow D6 TX, 4 green D7 RX.
Male and female halves carry the same colours in the same order on the crimp side, so the XIAO tail and the far tail match colour for colour through the joint.
XIAO tail: red -> 5V (14), black -> GND (13), yellow -> D6 (7), green -> D7 (8).
Far tail: red -> bus 5V row, black -> bus GND row, green -> bus TX row, yellow -> that sector's ESP RX pin (labels on the drawing). Female half on the far tail keeps 5V covered.

Design formulas - monopoles, triangle, baseplate (NOT a template, not to size)

Worked from parameters.csv: change a value there and re-run make_templates.py, and every number here follows.

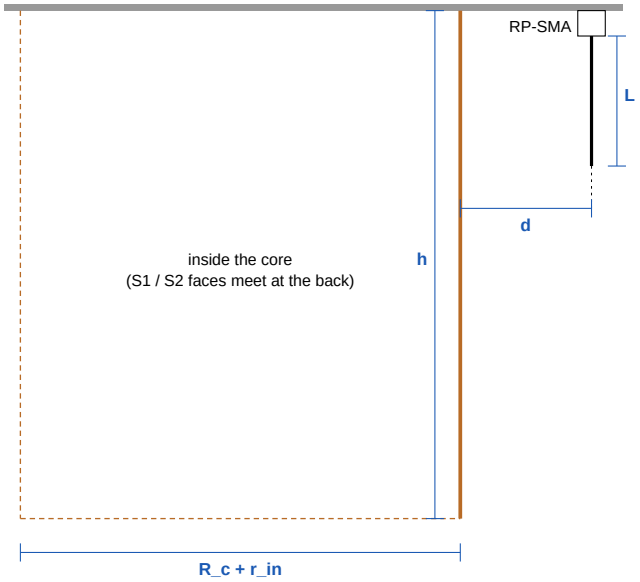
REFERENCE
both builds; no need to print
Pipe: pages 1, 2, 4, 5, 6 No pipe: pages 1, 3, 4, 6

TOP VIEW: baseplate (shield deck) + triangle core, S0 up, scale 0.7



Copper line = the triangle's outer (copper) faces. Circles = deck holes. Blue = the dimensions below.

SIDE VIEW: cut through the S0 axis, deck on top, scale 0.56



Whip drawn as a bare quarter-wave monopole; the deck is its ground plane, the face its reflector.

TRIANGLE CORE (3 back panels)

a = FACE_WIDTH (side of the triangle)	**120 mm**
r_in = a / (2 x sqrt 3) centre -> face (inradius)	**34.64 mm**
R_c = a / sqrt 3 = 2 x r_in centre -> corner (circumradius)	**69.28 mm**
height of the triangle = a x sqrt 3 / 2 = 3 x r_in	**103.9 mm**
back panel = a x h (h = FACE_H), make 3 panels meet at 60 deg inside / 120 deg between faces	**120 x 120 mm**
inner face = r_in - t (t = FACE_THICK)	**33.04 mm**
corner clearance = R - R_c must stay > 0: the corners sit inside the deck rim	**3.7 mm**
a / lambda (reflector width per sector) about one wavelength wide	**0.98**

BASEPLATE (shield deck)

R = DISC_DIAMETER / 2	**73 mm**
D hole: r_D = r_in + d at 0 / 120 / 240 deg	**65.64 mm**
D rim margin = R - r_D - HOLE_RPSMA / 2	**4.1 mm**
P hole: r_P = r_in + p (p = PATCH_OFFSET)	**41.64 mm**
P2: s = a / 2 - PATCH_END_INSET (along the face)	**25 mm**
P2: r = sqrt(r_P^2 + s^2), turned atan(s / r_P) CCW	**48.57 mm, 31.0 deg**
C hole: r_C = COAX_PASS_RADIUS (inside the core)	**20 mm**
M4: r_M4 = 2 x (r_in - t - NUT_AF / 2 - NUT_GAP) at the corners (60 / 180 / 300 deg): a hex nut in a 60 deg corner sits twice its flat-to-face distance from the corner, two flats on the faces	**59.08 mm**
stitch vias: STITCH_COUNT on STITCH_RADIUS	**36 on 71 mm**

MONOPOLES (D holes)

f = RF_FREQ (BLE band centre)	**2440 MHz**
lambda = c / f	**122.9 mm**
d = lambda / 4 (whip -> copper face) DUCK_SPACING = 31 mm (d / lambda = 0.252)	**30.7 mm**
why: out and back 2d = lambda / 2 (180) + copper flip (180) the reflected wave comes back in phase and adds to the forward wave	**360 deg**
L = lambda / 4 (bare monopole over the deck)	**30.7 mm**
rubber duck: lambda / 2 sleeve dipole element	**~61 mm**
backed length = h = FACE_H below the deck a whip longer than h hangs past the copper and loses some backing	**120 mm**
D hole to the far corner = R_c + r_in + d	**134.9 mm**

Units: mm and degrees, clockwise from S0 (top view) unless marked CCW. Names in capitals are parameters.csv rows.